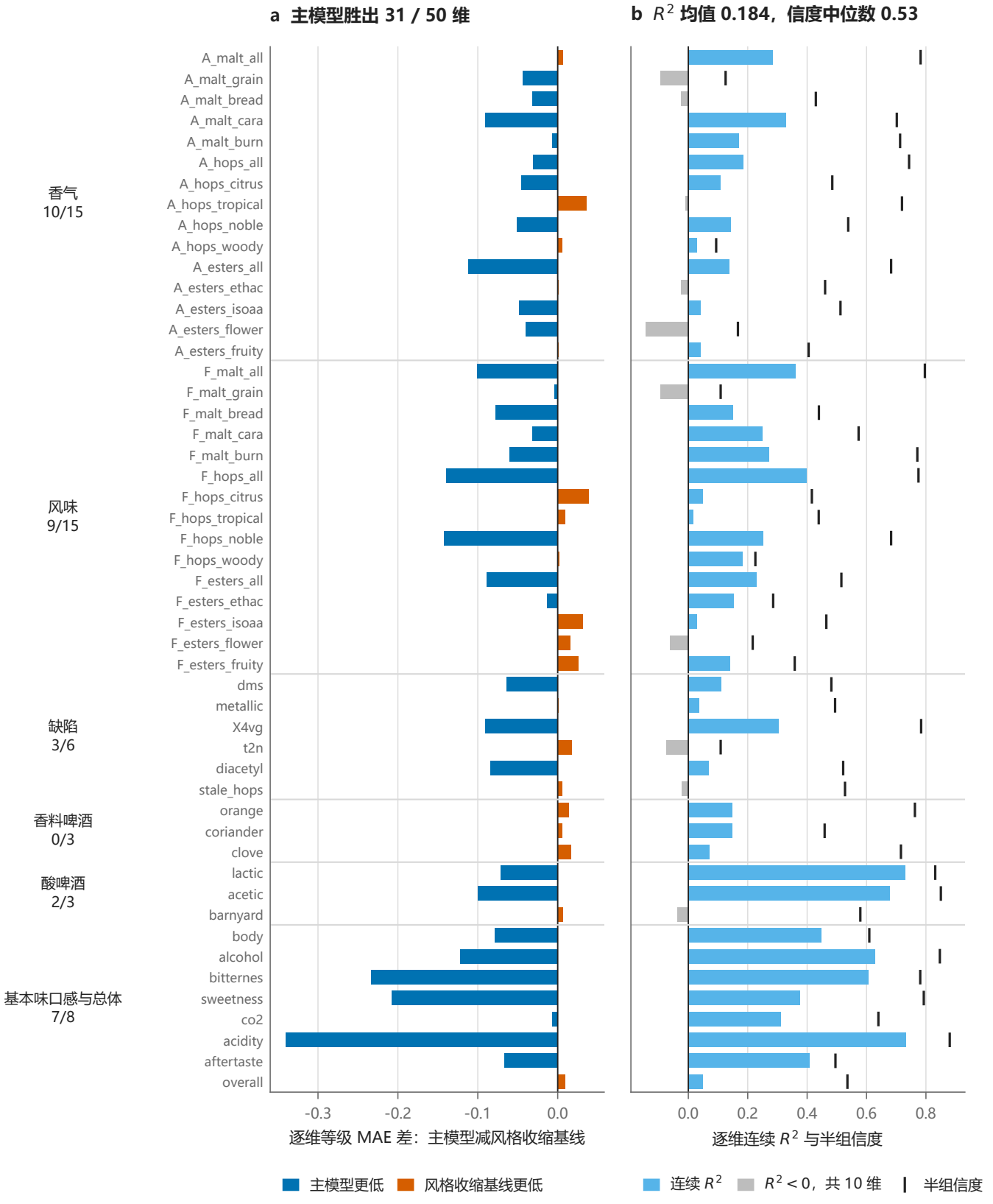

a 主模型胜出 31 / 50 维
b R^2 均值 0.184，信度中位数 0.53
香气
10/15
风味
9/15
缺陷
3/6
香料啤酒
0/3
酸啤酒
2/3
基本味口感与总体
7/8
A_malt_all
A_malt_grain
A_malt_bread
A_malt_cara
A_malt_burn
A_hops_all
A_hops_citrus
A_hops_tropical
A_hops_noble
A_hops_woody
A_esters_all
A_esters_ethac
A_esters_isoaa
A_esters_flower
A_esters_fruity
F_malt_all
F_malt_grain
F_malt_bread
F_malt_cara
F_malt_burn
F_hops_all
F_hops_citrus
F_hops_tropical
F_hops_noble
F_hops_woody
F_esters_all
F_esters_ethac
F_esters_isoaa
F_esters_flower
F_esters_fruity
dms
metallic
X4vg
t2n
diacetyl
stale_hops
orange
coriander
clove
lactic
acetic
barnyard
body
alcohol
bitternes
sweetness
co2
acidity
aftertaste
overall
-0.3
-0.2
-0.1
0.0
逐维等级 MAE 差：主模型减风格收缩基线
0.0
0.2
0.4
0.6
0.8
逐维连续 R^2 与半组信度
主模型更低
风格收缩基线更低
连续 R^2
$R^2 < 0$，共 10 维
半组信度